Sarwesh Rauniyar

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Education

Dartmouth College | Guarini School of Graduate and Advanced Studies — Hanover, NH
Ph.D. Computational Science and Modeling — September 2026 – June 2030

Johns Hopkins University | Whiting School of Engineering — Baltimore, MD
M.S.E. Applied Mathematics and Statistics | Major GPA: 3.98/4.00 — August 2024 – May 2025

University of Washington | College of Arts and Sciences — Seattle, WA
B.A. Mathematics | Dean's List Recipient — September 2018 – June 2022

Research Experience

IBM Research | Thomas J. Watson Research Center — June 2026 – September 2026
Research Scientist Intern: AI-Powered Automation — *Yorktown Heights, New York*

- Developed action-conditioned JEPA world models and evaluation signals for multivariate sensor environments, measuring long-horizon failure risk and system degradation under alternative maintenance interventions (Python, PyTorch).
- Designed and debugged horizon-event objectives for rare failures and intervention effects, raising failures caught from **66%** to **87%** and cutting false alarms by **5**×, creating decision-relevant signals beyond next-step prediction for comparing model behavior and counterfactual actions (PyTorch).

Lawrence Livermore National Laboratory | Physics Division — January 2026 – June 2026
Machine Learning Physics Researcher — *Livermore, California*

- Developed a Kolmogorov-Arnold Neural Network framework for nonlinear dimensionality reduction in scientific ML, compressing a high-dimensional **200**-term feature space by **92.5%** into **15** factors while lowering test error by **29%** (PyTorch).
- Applied mechanistic interpretability analyses, tracing how learned representations controlled equation terms and recovered dynamics to identify failure modes, debug spurious behavior, and improve model reliability (Python, PyTorch, NumPy).

Mercor | Research & Engineering — August 2025 – January 2026
AI Researcher — *Remote (contract)*

- Designed and ran an agentic evaluation framework for LLM reasoning models, comparing chain-of-thought and tree-search workflows and analyzing how agent design affected model behavior and failure modes; improved solve rates by **28%** (Python).
- Fine-tuned Llama 3.1 8B with RLHF post-training for reasoning tasks and scaled training across multi-node GPU clusters, achieving **3**× faster iteration and **22%** lower GPU memory use (PyTorch, Hugging Face, TRL, DeepSpeed, CUDA).
- Ran controlled ablations across training and inference-time agent configurations to separate the contributions of model training and agent workflows to reasoning performance (Python, PyTorch).

Johns Hopkins University | Applied Physics Laboratory — August 2024 – May 2025
Machine Learning Engineer — *Baltimore, MD*

- Engineered a multimodal autoregressive Transformer using custom attention mechanisms, positional encodings, and forward hooks to manipulate intermediate representations, reducing prediction error by **15%** in physical systems (Python, PyTorch).
- Scaled a pretraining framework for deep neural networks with distributed GPU orchestration, reducing iteration time by **60%** and enabling rapid prototyping across **5+ model architectures** (Python, PyTorch DDP, JAX).

Fred Hutchinson Cancer Center | Translational Science Division — September 2022 – June 2024
Research Assistant — *Seattle, WA*

- Developed an end-to-end computer vision pipeline for biomedical screening, developing custom CNN architectures that achieved **89%** classification accuracy on **10k+ medical images** and enabled high-throughput cancer screening (Python, OpenCV).
- Engineered a scalable transfer learning framework integrating multi-source imaging datasets into ResNet architectures, reducing required training data by **40%** and lowering annotation costs while accelerating deployment timelines (Python, PyTorch).
- https://github.com/DeyLab/Orouskhani_Rauniyar_et_al_2025

Publications & Preprints

[1] **When Reasoning Erasure Stops Too Early** — **ICLR 2027 (Under Review)**
Rauniyar S.

[2] **Training Schedule Order Selects Isospectral Matrix Completions** — **ICLR 2027 (Under Review)**
Rauniyar S.

[3] **Statistical Risk Heads for Failure Warning Across Time-Series Forecasters** — **ICLR 2027 (Under Review)**
Rauniyar S., Jayaraman S., Bhamidipaty A., Fokoue A.

[4] **Model Collapse is a Singular Complexity Trajectory** — **NeurIPS 2026**
Rauniyar S.

[5] **Jacobian Aligned Random Forests** — **ICLR 2026**
Rauniyar S.

[6] **Selective Cost-Aware Random Forests for Unreliable Data** — **NeurIPS 2025**
Rauniyar S.

[7] **When Curvature Beats Dimension: Euclidean Limits and Hyperbolic Design Rules for Trees** — **NeurIPS 2025**
Rauniyar S.

[8] **Deep Learning Imaging Analysis to Identify Bacterial Metabolic States Associated with Carcinogen Production** — **Discover Imaging 2025**
Orouskhani M., **Rauniyar S.**, Morella N., *et al.*

Intellectual Property

[1] **"Smoothed Early-Warning Failure Risk Prediction for Predictive Maintenance."** — September 2026
Rauniyar, S., Jayaraman, S., Bhamidipaty, A., Fokoue-Nkoutche, A. B.
IBM Invention Disclosure P202603525, under patent review

Selected Fellowships & Recognition

- **Anthropic AI Safety Fellows Program — Finalist** — *2026*
- **MATS Research Fellowship — Finalist, Empirical Stream** — *2026*
  Cozmin Ududec (UK AI Security Institute) and Apollo Research (Science of Scheming)
- **MATS Research Fellowship — Finalist, Founding & Field-Building Stream** — *2026*
  Geoff Ralston (Safe AI Fund; former President, Y Combinator) and Charlie Petty (Stripe)
- **Cambridge Boston Alignment Initiative (CBAI) Fellowship — Finalist** — *2026*
- **Existential Risk Alliance (ERA) AI Fellowship — Finalist** — *2026*
- **Princeton Prospective Ph.D. Preview (P3) — Scholar** — *2026*
  Accepted as one of 50 scholars selected from 500 applicants
- **ICLR 2026 Financial Assistance Award** — *March 2026*
  $2,100
- **National GEM Ph.D. Engineering and Science Fellowship** — *December 2025*
  $20,000 with IBM Research Sponsorship
- **NeurIPS 2025 Travel Award** — *November 2025*
  $500
- **Johns Hopkins Technology Ventures Entrepreneurship Grant** — *December 2024; May 2025*
  $5,500
- **Johns Hopkins Applied Math and Statistics Departmental Scholarship** — *August 2024*
  $11,300
- **Society of American Military Engineers Seattle Post Scholarship** — *September 2018 – June 2022*
  $12,000
- **NSF Research Experience for Undergraduates (REU) Award, Arizona State University** — *June 2021*
  $7,000 through the Quantitative Research in the Life and Social Sciences Program (QRLSSP)

Technical Skills

Languages: Python, C/C++, Java, Bash, SQL, JavaScript, R, MATLAB.
ML Frameworks: PyTorch (DDP, DeepSpeed, FSDP), JAX, TensorFlow, Hugging Face, CUDA.

Professional Service & Memberships

- **Reviewer**, International Conference on Learning Representations (ICLR), 2027
- **Reviewer**, Conference on Neural Information Processing Systems (NeurIPS), 2026
- **Member**, Society for Industrial and Applied Mathematics (SIAM), Johns Hopkins Student Chapter